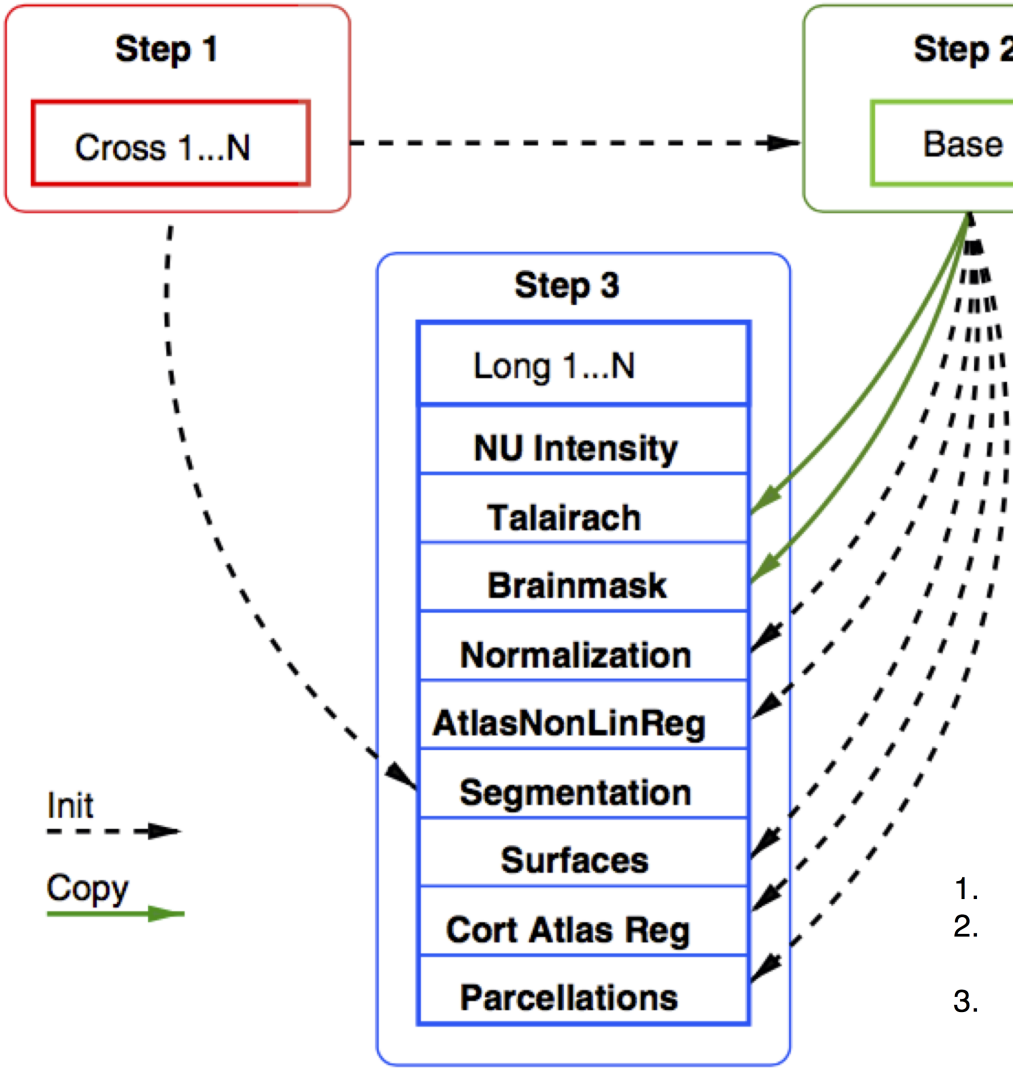
The ANTs longitudinal cortical thickness pipeline

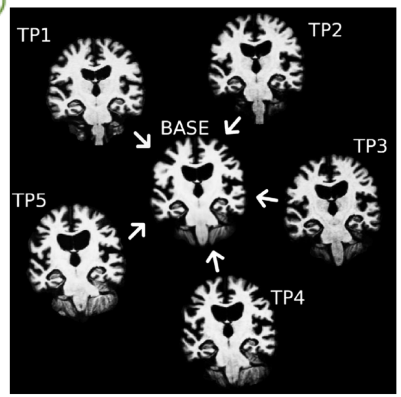
N. Tustison et al.

UVa/UCI

Longitudinal cortical structural changes from MRI

FreeSurfer longitudinal pipeline



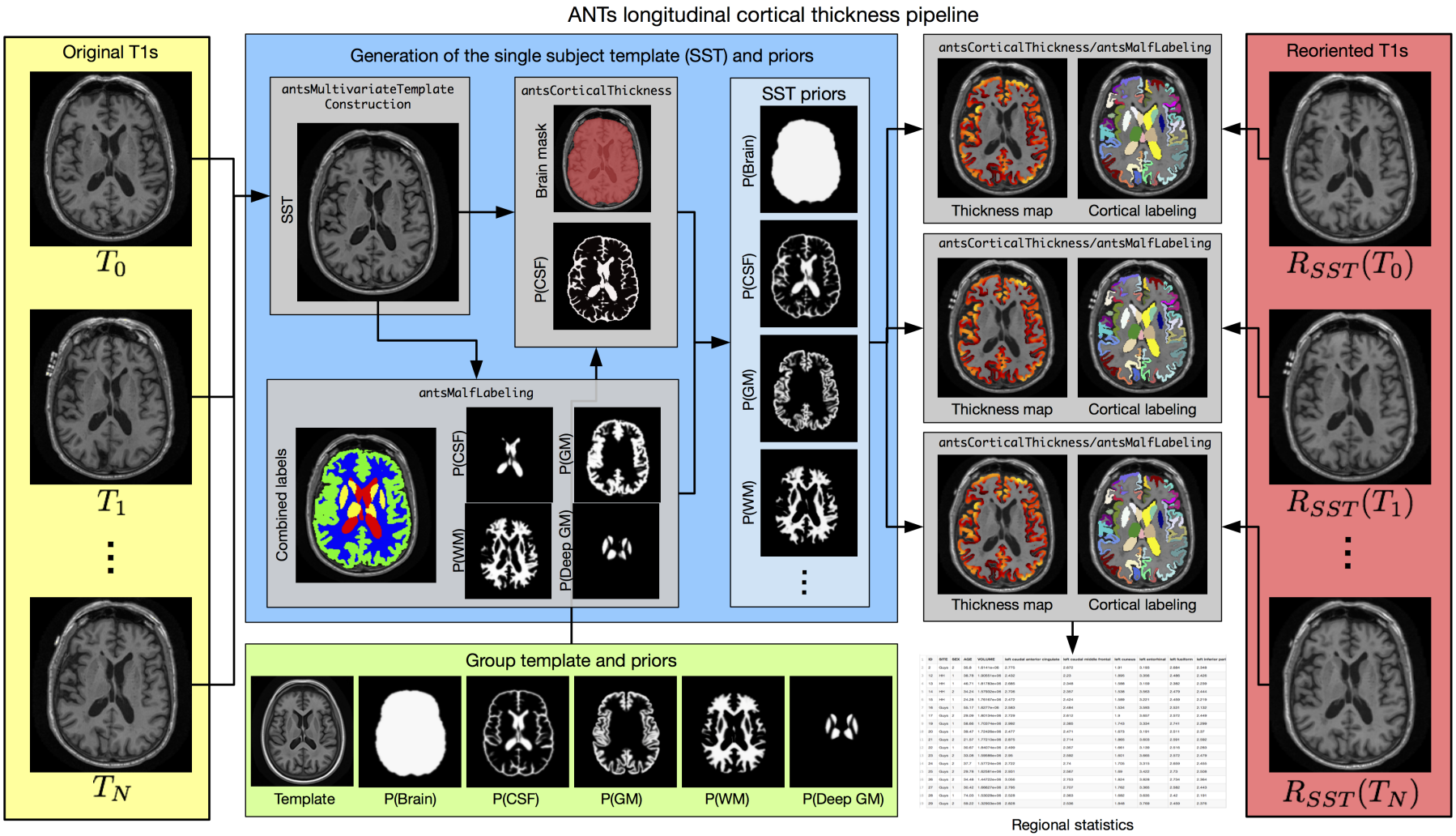


Median template ("Base")

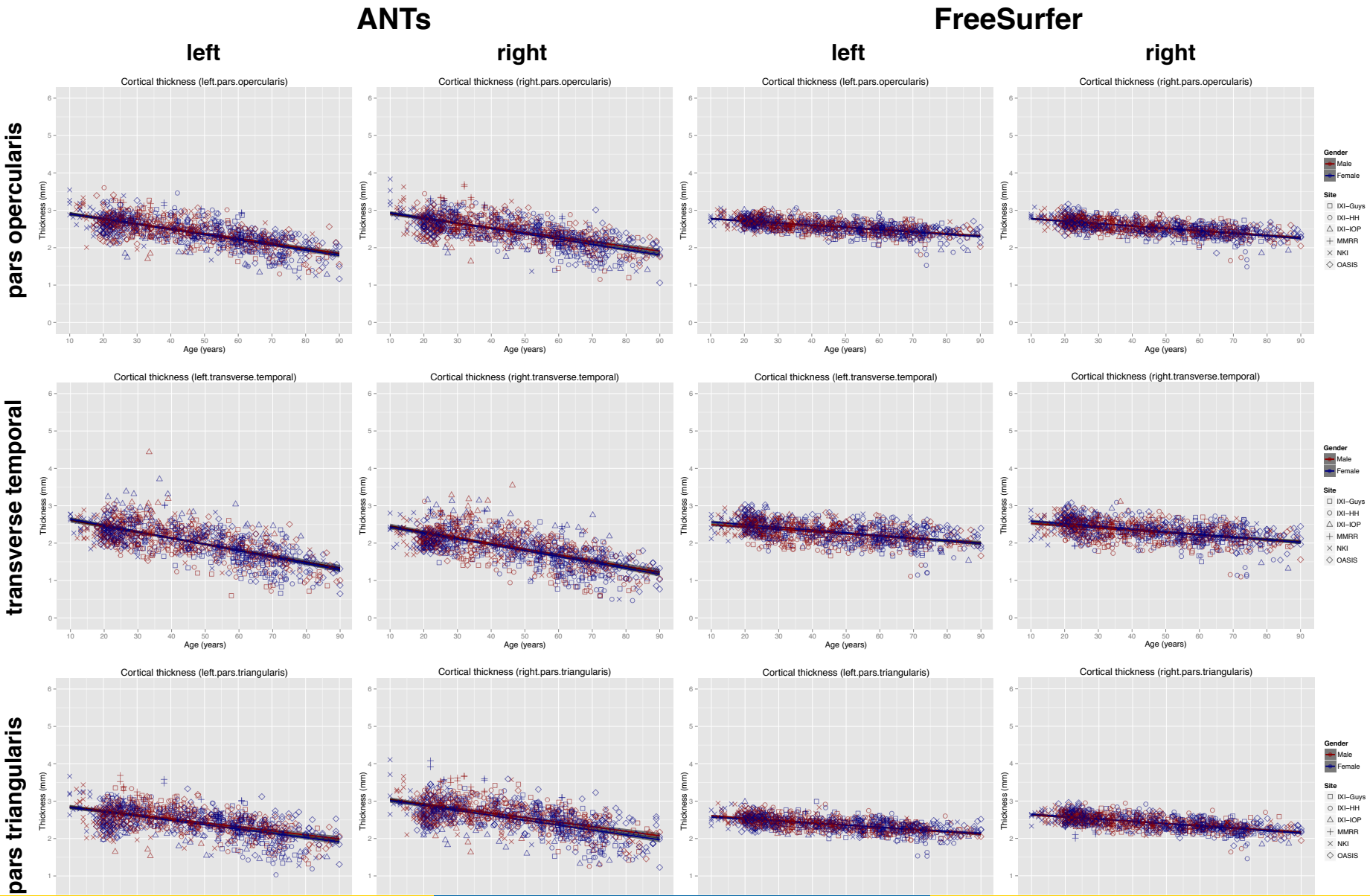
FreeSurfer long. steps

1. FS processing of each time point.
2. Median template construction and FS processing.
3. FS processing of each rotated time point with "Base" initialization.

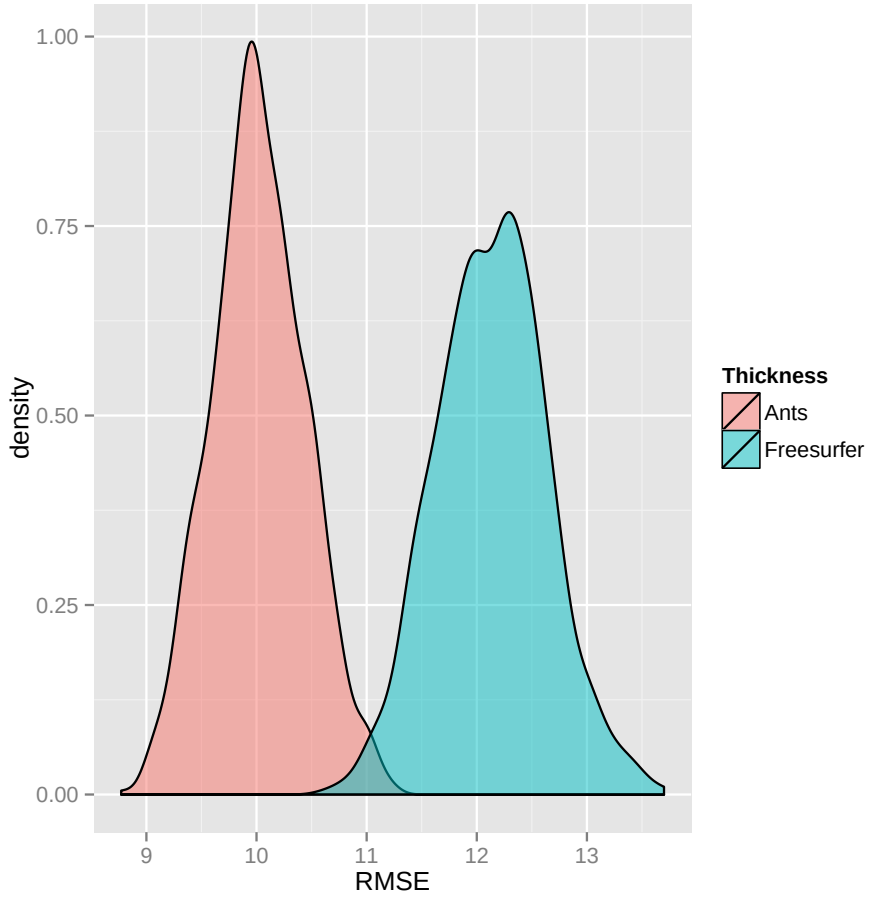
ANTs longitudinal pipeline



Cross-sectional comparison

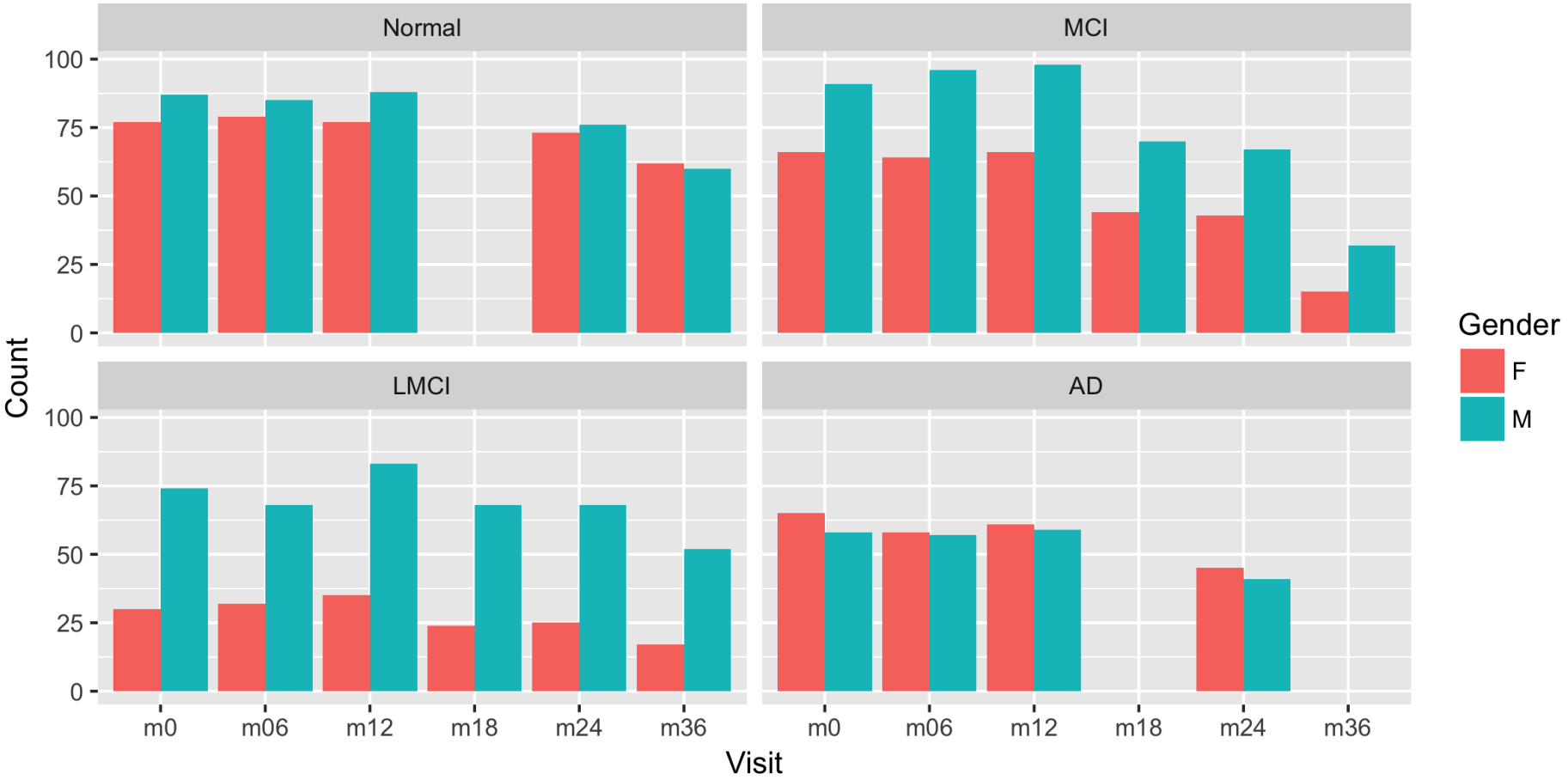


Cross-sectional evaluation (age prediction)

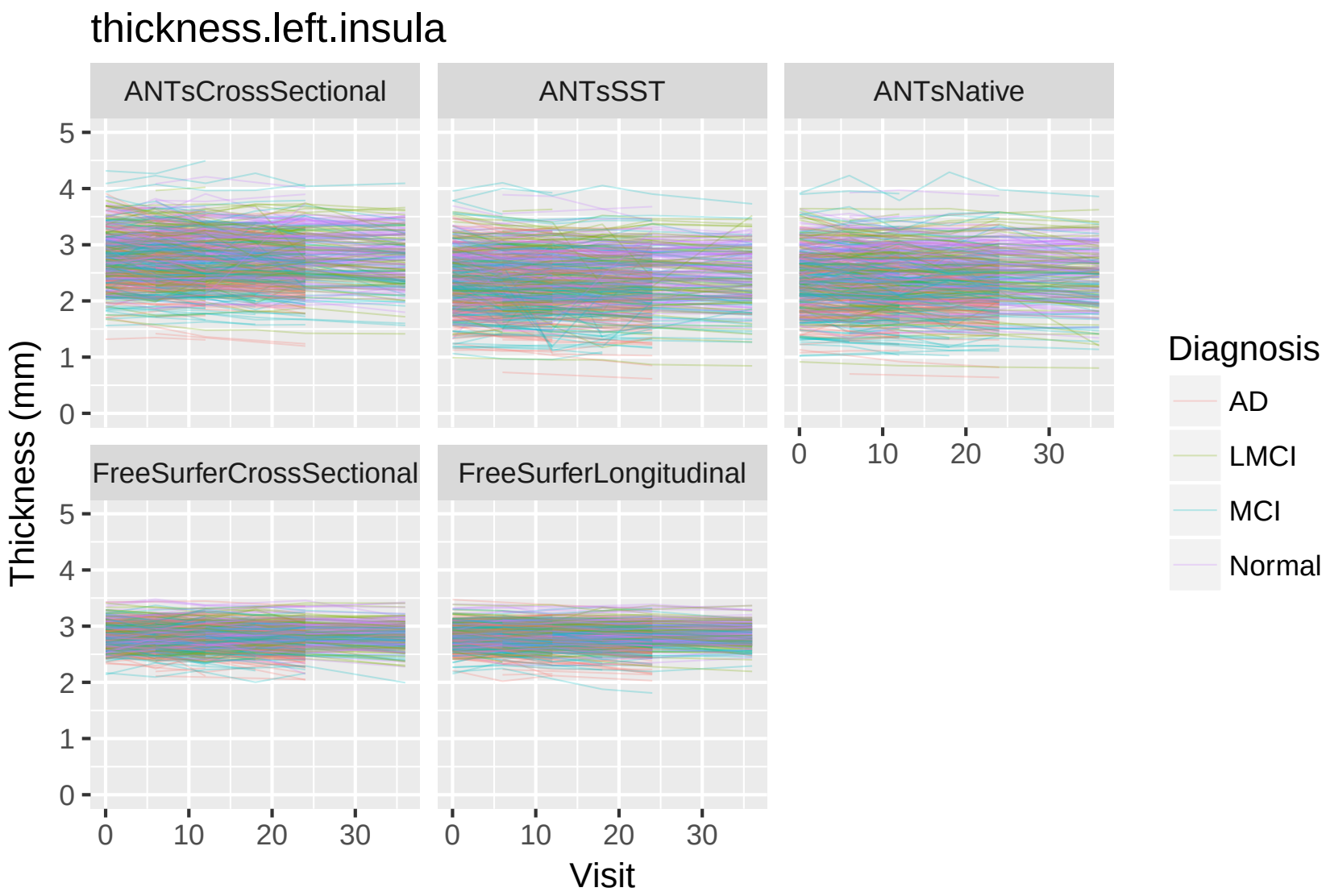


Repeatability: $ICC_{FS} = 0.97, ICC_{ANTs} = 0.98$

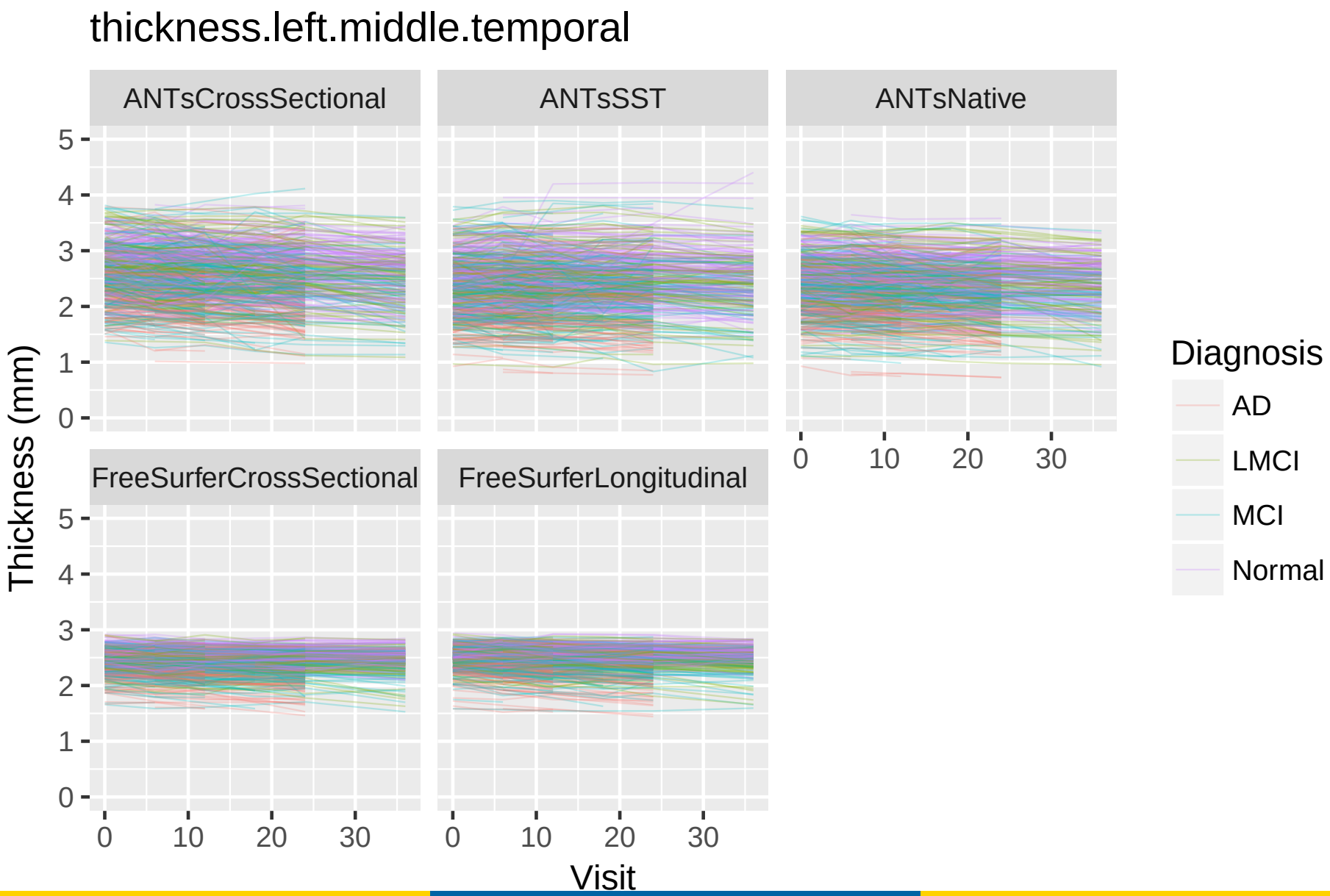
Longitudinal comparison (ADNI-1)



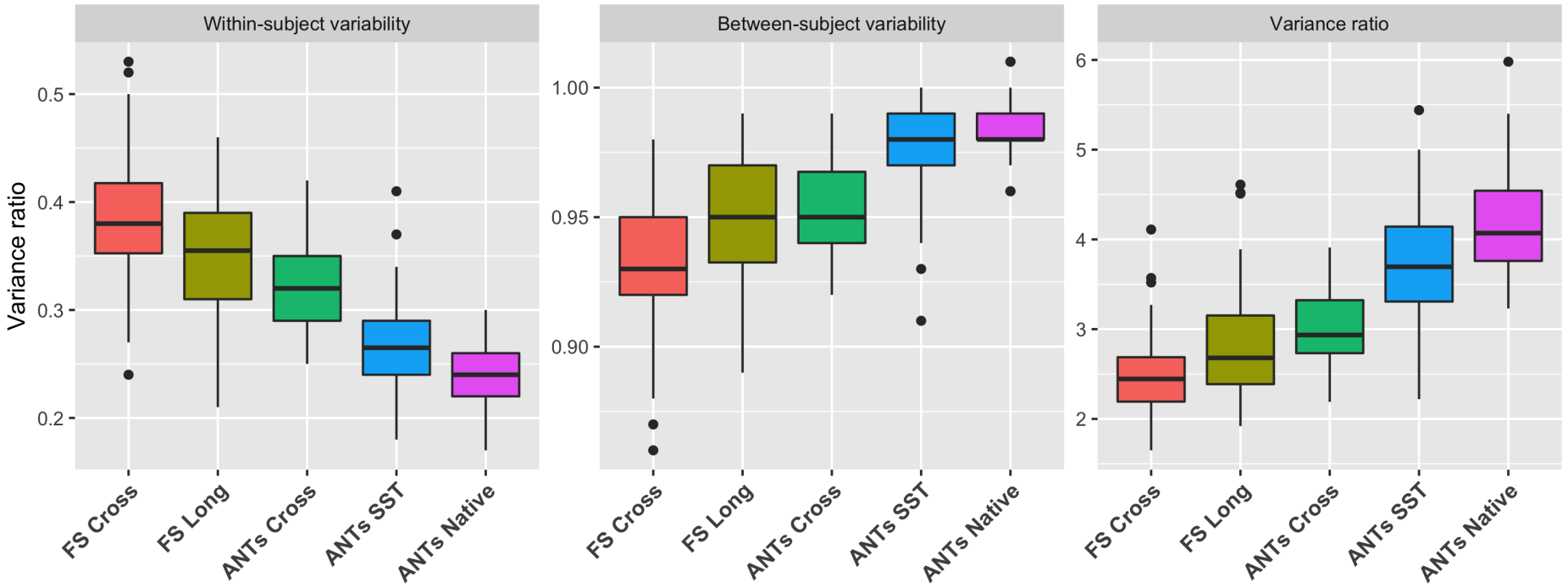
Spaghetti plots (left insula)



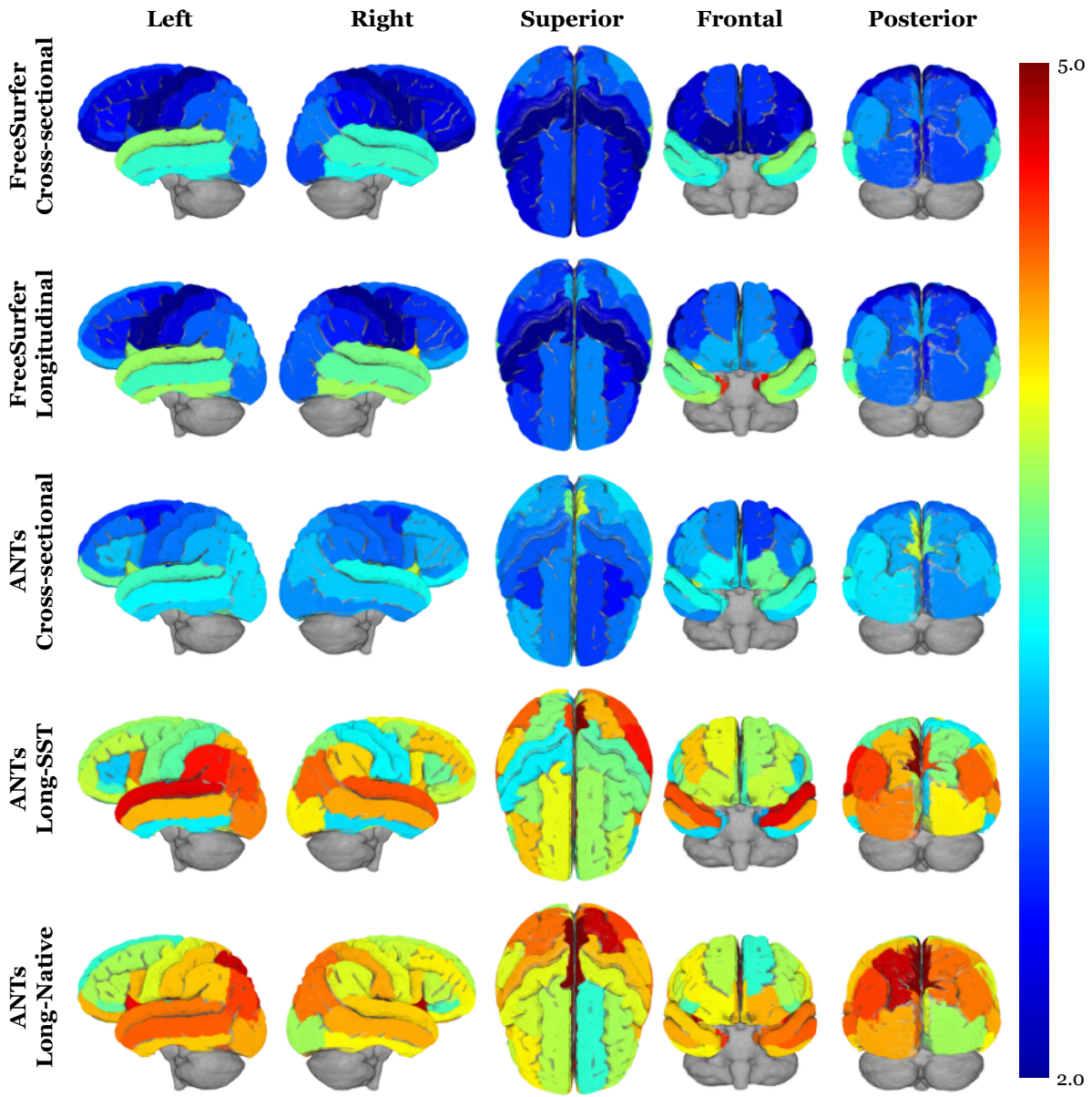
Spaghetti plots (left middle temporal)



Within-subject and between-subject variability



Regional comparisons



Issues in the EC

Xie et al., *Accounting for the Confound of Meninges in Segmenting Entorhinal and Perirhinal Cortices in T1-Weighted MRI*, MICCAI 2016.

